

Equal-Weight vs Risk-Parity Comparison Report

Report Scope

- **Source:** ew_rp_comparison.txt
- **Calculation window:** 10y (window_months=120)
- **Analysis end date:** 2026-02-28
- **Delta rule:** EW - RP

Executive Summary

- Equal-Weight (EW) shows higher return metrics than Risk-Parity (RP), with higher total and downside volatility.
- Risk-Parity (RP) demonstrates lower volatility and better tail risk values (ES_95, ES_99) in absolute terms.
- Both portfolios fail the same stress check (FAIL_ROLE_EQUITY_SHOCK) while remaining marked as portfolio_valid=True.

Core Metrics Comparison

Metric	EW	RP	Delta (EW - RP)
CAGR	14.95%	11.21%	3.70%
Annual Volatility	12.50%	9.84%	2.70%
Max Drawdown	-19.93%	-18.75%	-1.20%
Sharpe	1.007	0.910	0.097
Sortino	1.722	1.471	0.251
Beta (Portfolio)	0.745	0.602	0.143
Treynor	0.169	0.149	0.020
Correlation to Base	0.891	0.915	-0.024
Downside Deviation (Annual)	7.30%	6.10%	1.20%
Skewness	-0.416	-0.516	0.100
Kurtosis	0.647	0.781	-0.134
ES 95	-7.30%	-5.90%	-1.40%
ES 99	-8.80%	-7.40%	-1.40%
EEE 10%	96.069	74.197	21.872
Time to Recovery (months)	—	—	—

Rolling Metrics

Rolling Sharpe 36m (last / mean / p10 / p90)

- **EW:** 1.798 / 0.724 / 0.311 / 1.140
- **RP:** 1.681 / 0.672 / 0.190 / 1.128
- **Delta (EW - RP):** 0.117 / 0.052 / 0.121 / 0.012

Rolling Sharpe 12m (last / mean / p10 / p90)

- **EW:** 3.290 / 1.040 / -0.403 / 2.537
- **RP:** 3.712 / 1.042 / -0.383 / 2.464
- **Delta (EW - RP):** -0.422 / -0.002 / -0.020 / 0.073

Rolling Volatility 12m (last / mean / p10 / p90)

- **EW:** 10.20% / 11.80% / 7.40% / 18.20%
- **RP:** 6.00% / 9.30% / 5.00% / 15.20%
- **Delta (EW - RP):** 4.20% / 2.50% / 2.40% / 3.00%

Volatility Stability

- **vol_of_vol:** EW = 0.040 | RP = 0.037 | Delta = 0.003
- **rel_vol_of_vol:** EW = 0.339 | RP = 0.396 | Delta = -0.057

Risk Contribution by Blocks (RC_vol)

- **Duration:** EW = 1.00% | RP = 3.20% | Delta = -2.20%
- **Growth:** EW = 89.60% | RP = 82.40% | Delta = 7.20%
- **Inflation:** EW = 9.30% | RP = 14.40% | Delta = -5.10%
- **Liquidity:** EW = 0.00% | RP = 0.00% | Delta = 0.00%

Risk Contribution by Asset (RC_vol)

- ARMY.PA: EW = 0.60% | RP = 0.10% | Delta = 0.50%
- BBJP: EW = 4.00% | RP = 4.30% | Delta = -0.30%
- BIL: EW = -0.00% | RP = 0.00% | Delta = -0.00%
- BND: EW = 1.00% | RP = 3.20% | Delta = -2.20%
- CIBR: EW = 5.30% | RP = 5.10% | Delta = 0.20%
- COPX: EW = 11.40% | RP = 4.00% | Delta = 7.40%
- GLD: EW = 2.20% | RP = 2.20% | Delta = 0.00%
- ITA: EW = 6.30% | RP = 4.30% | Delta = 2.00%
- QQQ: EW = 6.00% | RP = 8.00% | Delta = -2.00%
- ROBO: EW = 7.80% | RP = 9.70% | Delta = -1.90%
- SCHD: EW = 4.90% | RP = 6.60% | Delta = -1.70%
- SCHP: EW = 1.10% | RP = 3.50% | Delta = -2.40%
- SLV: EW = 6.00% | RP = 8.70% | Delta = -2.70%
- SMH: EW = 8.40% | RP = 4.30% | Delta = 4.10%
- URA: EW = 9.30% | RP = 1.60% | Delta = 7.70%
- VDC: EW = 3.40% | RP = 5.40% | Delta = -2.00%
- VGK: EW = 5.80% | RP = 6.90% | Delta = -1.10%
- VOO: EW = 5.50% | RP = 9.50% | Delta = -4.00%
- VT: EW = 5.60% | RP = 7.70% | Delta = -2.10%
- VWO: EW = 5.20% | RP = 5.10% | Delta = 0.10%

Top-5 RC_vol Assets (as reported)

EW Top-5 by RC_vol

- COPX: EW = 11.40%
- URA: EW = 9.30%
- SMH: EW = 8.40%
- ROBO: EW = 7.80%
- ITA: EW = 6.30%

RP Top-5 by RC_vol

- ROBO: RP = 9.70%
- VOO: RP = 9.50%
- SLV: RP = 8.70%
- QQQ: RP = 8.00%
- VT: RP = 7.70%

Stress Status

- **EW stress:** FAIL_STRESS (FAIL_ROLE_EQUITY_SHOCK) | portfolio_valid = True
- **RP stress:** FAIL_STRESS (FAIL_ROLE_EQUITY_SHOCK) | portfolio_valid = True

Conclusion / Key Takeaways

- EW outperforms RP on return-focused ratios (CAGR, Sharpe, Sortino, Treynor) in this 10-year window.
- RP keeps lower overall and downside volatility, and better expected shortfall values.
- Both portfolios share the same stress failure status, so comparative differences do not remove stress vulnerability in the reported scenario.